

FIELD THEORY PART-II

Absolute-Zero to Semester-Ready Roadmap

For B.E. Electrical Engineering Second Year

Based on Sir's Syllabus + PYQ Pattern + Available Materials

Main Aim of This Roadmap

This roadmap is made for a student who starts from almost **absolute zero**. The target is not only to understand Field Theory Part-II, but also to write a clean semester answer-script efficiently.

The whole Part-II preparation is reduced to one chain:

Vector Calculus → Magnetostatics → Maxwell Equations → EM Waves → PYQ Writing Practice

After completing all phases in this document, no extra phase should be required for Part-II semester preparation.

Contents

1	Final Part-II Syllabus Understanding	2
2	High-Priority PYQ Map	4
3	Phase-by-Phase Roadmap	5
3.1	Phase 0: Absolute Zero Orientation	5
3.2	Phase 1: Basic Vector Language and Coordinates	5
3.3	Phase 2: Curl Mastery	6
3.4	Phase 3: Divergence Mastery	6
3.5	Phase 4: Stokes' Theorem and Ampere's Law	7
3.6	Phase 5: Magnetostatics Core	8
3.7	Phase 6: Magnetic Boundary Conditions	8
3.8	Phase 7: Time-Varying Electromagnetic Fields	9
3.9	Phase 8: EM Wave Equation and Plane Wave	10
3.10	Phase 9: Poynting, Polarization, Skin Effect and Concepts	11
3.11	Phase 10: PYQ Answer-Writing and Final Revision	12
4	Recommended Study Time Distribution	14
5	Final Answer-Writing Strategy for Semester	15
6	Final Phase Completion Checklist	17

1 Final Part–II Syllabus Understanding

What Part–II Actually Contains

Your Field Theory Part–II mainly contains:

1. Mathematical foundation:

gradient, divergence, curl, Stokes' theorem, coordinate systems

2. Magnetostatics:

Biot–Savart law, Ampere's law, magnetic vector potential, magnetic boundary condition

3. Time-varying electromagnetic field:

displacement current, modified Ampere law, Faraday law, Maxwell equations

4. Electromagnetic waves:

wave equation, plane wave solution, velocity, Poynting vector, polarization

5. Short notes:

$$\nabla \cdot \mathbf{J} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{H} = \mathbf{J}, \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0$$

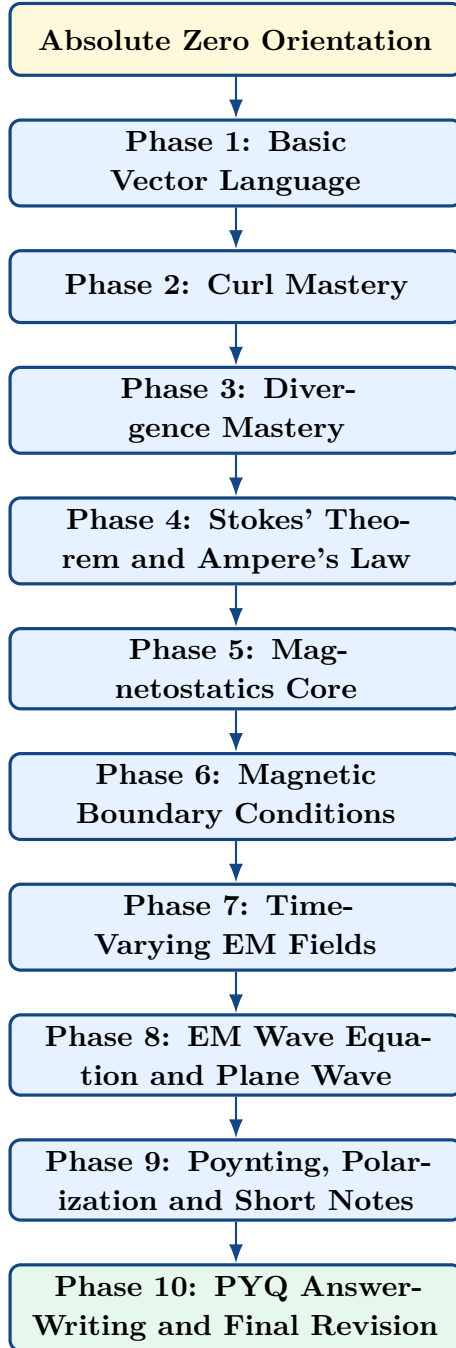


Figure 1: Complete phase-wise preparation flow for Field Theory Part-II.

2 High-Priority PYQ Map

Repeated Question Pattern

From the past papers, the following topics are the most repeated and safest for semester preparation:

Curl $\rightarrow$ Stokes $\rightarrow$ Ampere $\rightarrow$ Modified Ampere $\rightarrow$ Faraday $\rightarrow$ EM Wave $\rightarrow$ Poynting $\rightarrow$ Polarization

No.	Topic	Priority	Why Important
1	Definition and derivation of curl in Cartesian coordinates	Very High	Asked repeatedly as long answer.
2	Physical significance of curl	Very High	Often attached with curl derivation or short note.
3	Stokes' theorem	Very High	Direct question and also used to derive Ampere's law.
4	Divergence of $\mathbf{B}$ and $\nabla \cdot \mathbf{B} = 0$	Very High	Repeated as derivation and explanation.
5	Divergence of $\mathbf{J}$ and $\nabla \cdot \mathbf{J} = 0$	Very High	Common short/medium answer.
6	Ampere's law $\nabla \times \mathbf{H} = \mathbf{J}$	Very High	Repeated every year in some form.
7	Magnetic boundary condition	Very High	Long-answer repeated topic.
8	Biot-Savart law loop numerical	High	Numericals appear repeatedly.
9	Modified Ampere law and displacement current	Very High	Direct repeated long answer.
10	Faraday law $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$	Very High	Direct repeated long answer.
11	EM wave equation from Maxwell equations	Very High	Long-answer guaranteed type.
12	Plane wave solution and wave velocity	Very High	Often attached to wave equation.
13	Poynting vector and Poynting theorem	Very High	Common long answer and short note.
14	Polarization: linear, circular, elliptical	Very High	Repeated short/long topic.
15	Magnetic vector potential	Medium-High	Appeared in older paper; prepare as backup.
16	Skin effect, antenna, waveguide concept	Medium	Sir's syllabus topic; prepare short notes.

3 Phase-by-Phase Roadmap

3.1 Phase 0: Absolute Zero Orientation

Goal of Phase 0

This phase is only for understanding what the subject is about. Do not start derivations immediately.

Field Theory studies how electric and magnetic quantities vary in space and time.

In Part-II, the main story is:

Current produces magnetic field → Changing electric field also produces magnetic field → Changing mag

What to know before Phase 1

Memorise the basic field quantities:

Quantity	Meaning	Unit
E	Electric field intensity	V/m
D	Electric flux density	C/m^2
H	Magnetic field intensity	A/m
B	Magnetic flux density	T
J	Current density	A/m^2
S	Poynting vector	W/m^2

3.2 Phase 1: Basic Vector Language and Coordinates

Goal of Phase 1

Before curl, divergence, Stokes, and Maxwell equations, you must understand vector language.

Study these in this order:

1. Unit vectors:

$$\hat{\mathbf{x}}, \quad \hat{\mathbf{y}}, \quad \hat{\mathbf{z}}$$

2. Vector field:

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

3. Differential length:

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$$

4. Differential surface:

$$d\mathbf{S} = dS \hat{\mathbf{n}}$$

5. Differential volume:

$$dV = dx \, dy \, dz$$

Phase 1 Output

After this phase, you should be able to understand:

$$\mathbf{A} \cdot d\mathbf{l}, \quad \mathbf{A} \cdot d\mathbf{S}, \quad \oint_C \mathbf{A} \cdot d\mathbf{l}, \quad \oint_S \mathbf{A} \cdot d\mathbf{S}$$

Without this, curl and divergence will feel confusing.

3.3 Phase 2: Curl Mastery

Goal of Phase 2

Curl is the most repeated Part-II topic. You must prepare it as a complete 8-mark answer.

Study these blocks:

1. Fundamental definition:

$$(\nabla \times \mathbf{A}) \cdot \hat{\mathbf{n}} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \mathbf{A} \cdot d\mathbf{l}$$

2. Derivation of z -component using small rectangular loop.
3. Final Cartesian form:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

4. Physical significance:

Curl means local circulation or rotational tendency.

Phase 2 Output

You should be able to write:

- definition of curl,
- derivation of curl in Cartesian coordinates,
- determinant formula,
- physical significance,
- simple diagram of small rectangular loop.

3.4 Phase 3: Divergence Mastery

Goal of Phase 3

Divergence is needed for $\nabla \cdot \mathbf{B} = 0$, $\nabla \cdot \mathbf{J} = 0$, and continuity equation.

Study these blocks:

1. Fundamental definition:

$$\nabla \cdot \mathbf{A} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oint_S \mathbf{A} \cdot d\mathbf{S}$$

2. Cartesian expression:

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

3. Magnetic flux law:

$$\nabla \cdot \mathbf{B} = 0$$

4. Steady current condition:

$$\nabla \cdot \mathbf{J} = 0$$

5. Identity:

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

Phase 3 Output

You should be able to write short notes on:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \cdot \mathbf{J} = 0, \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0$$

3.5 Phase 4: Stokes' Theorem and Ampere's Law

Goal of Phase 4

This phase connects mathematics to magnetostatics.

First study Stokes' theorem:

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Meaning:

$$\text{closed line integral} = \text{surface integral of curl}$$

Then study Ampere's law:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

where

$$I_{\text{enc}} = \iint_S \mathbf{J} \cdot d\mathbf{S}$$

Using Stokes' theorem:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

Phase 4 Output

You should be able to answer:

- Establish Stokes' theorem.
- Establish $\nabla \times \mathbf{H} = \mathbf{J}$.
- Explain its physical significance.

3.6 Phase 5: Magnetostatics Core

Goal of Phase 5

This phase covers magnetic fields due to steady current.

Study these topics:

1. Biot–Savart law:

$$d\mathbf{B} = \frac{\mu I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}$$

2. Magnetic field intensity:

$$d\mathbf{H} = \frac{I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}$$

3. Magnetic vector potential:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

4. Rectangular or square loop numerical from Biot–Savart law.

5. Magnetic energy density:

$$w_m = \frac{B^2}{2\mu} = \frac{1}{2}\mu H^2$$

Most Important Numerical Area

Practise at least two cases:

- rectangular loop on its central axis,
- square loop in xz -plane with point on y -axis.

These numericals are not to be memorised blindly. You must know the diagram, axis, direction of current, and final formula.

3.7 Phase 6: Magnetic Boundary Conditions

Goal of Phase 6

This is a repeated long-answer topic.

Study two boundary conditions:

Tangential component of $\mathbf{H}$:

$$H_{1t} = H_{2t}$$

if there is no surface current.

Normal component of $\mathbf{B}$:

$$B_{1n} = B_{2n}$$

Using:

$$\mathbf{B}_1 = \mu_1 \mathbf{H}_1, \quad \mathbf{B}_2 = \mu_2 \mathbf{H}_2$$

we obtain:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

Phase 6 Output

You should be able to write the full answer with:

- interface diagram,
- tangential H condition,
- normal B condition,
- magnetic refraction relation,
- one application: transformer core or magnetic shielding.

3.8 Phase 7: Time-Varying Electromagnetic Fields

Goal of Phase 7

This phase explains how changing fields generate each other.

Study these in order:

1. Displacement current

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

2. Modified Ampere law

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

3. Faraday law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

4. Continuity equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Phase 7 Output

You should be able to write:

- modified Ampere law with capacitor diagram,
- displacement current definition,
- Faraday law with circular electric field diagram,
- physical meaning of negative sign.

3.9 Phase 8: EM Wave Equation and Plane Wave

Goal of Phase 8

This is the biggest long-answer area.

Start from Maxwell's equations:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}$$

For a source-free conducting medium:

$$\nabla^2 \mathbf{E} = \mu \sigma \frac{\partial \mathbf{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{H} = \mu \sigma \frac{\partial \mathbf{H}}{\partial t} + \mu \epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

In free space:

$$\sigma = 0, \quad \mu = \mu_0, \quad \epsilon = \epsilon_0$$

So,

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

For plane wave along $+x$:

$$E_y(x, t) = E_0 \cos(\omega t - \beta x)$$

or

$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

where

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}$$

and

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s}$$

Phase 8 Output

You should be able to write a 10-mark answer on:

- EM wave equation from Maxwell equations,
- free-space wave equation,
- plane wave assumption,
- analytical solution,
- wave velocity,
- wave propagation diagram.

3.10 Phase 9: Poynting, Polarization, Skin Effect and Concepts

Goal of Phase 9

This phase covers repeated short notes and medium long answers.

1. Poynting vector

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Unit:

$$W/m^2$$

2. Poynting theorem

$$\nabla \cdot \mathbf{S} = -\frac{\partial u_{em}}{\partial t} - \mathbf{E} \cdot \mathbf{J}$$

where

$$u_{em} = \frac{1}{2}\epsilon E^2 + \frac{1}{2}\mu H^2$$

3. Polarization

Linear, Circular, Elliptical

4. Skin effect

Skin effect means alternating current tends to concentrate near the surface of a conductor at high frequency.

5. Antenna concept

An antenna converts electrical signals into electromagnetic waves and vice versa.

6. Waveguide concept

A waveguide guides electromagnetic waves from one point to another with low radiation loss.

Phase 9 Output

You should be able to write short notes on:

Poynting vector, Poynting theorem, polarization, skin effect, antenna, waveguide

3.11 Phase 10: PYQ Answer-Writing and Final Revision

Goal of Phase 10

This is the final conversion phase: from knowing concepts to writing marks-scoring semester answers.

For every long answer, follow this writing pattern:

Definition → Diagram → Mathematical derivation → Physical meaning → Final boxed result

For every short note, follow:

Definition → Formula → Meaning → Application

Practise past years in this order:

1. 2025 Part-II.
2. 2024 Part-II.
3. 2023 Part-II.
4. 2022 Part-II.

Do not only read answers. Write them by hand at least once.

Phase 10 Output

After this phase, you should be ready to write the semester paper efficiently.

You should be able to attempt any three long questions confidently and use short notes as backup.

4 Recommended Study Time Distribution

If You Have Very Less Time

Give more time to repeated PYQ topics. The safest scoring order is:

Curl $\rightarrow$ Stokes $\rightarrow$ Ampere $\rightarrow$ Modified Ampere $\rightarrow$ Faraday $\rightarrow$ EM Wave $\rightarrow$ Poynting $\rightarrow$ Polarization

Day	Main Work	Time	Output
1	Vector basics, curl derivation, physical significance	4–5 hrs	One full curl answer ready.
2	Divergence, $\nabla \cdot \mathbf{B} = 0$, $\nabla \cdot \mathbf{J} = 0$, Stokes theorem	4–5 hrs	Three short/medium answers ready.
3	Ampere law, Biot–Savart, magnetic vector potential, loop numerical	5–6 hrs	Magnetostatics block ready.
4	Magnetic boundary condition, displacement current, modified Ampere law, Faraday law	5–6 hrs	Two long answers ready.
5	EM wave equation, plane wave solution, velocity	5–6 hrs	Most important long answer ready.
6	Poynting theorem, polarization, skin effect, antenna, waveguide	4–5 hrs	Short-note section ready.
7	Full PYQ writing practice and formula revision	5–6 hrs	Final exam simulation.

5 Final Answer-Writing Strategy for Semester

How to Choose Questions in Exam

In Part–II, choose questions by this priority:

1. Choose EM wave equation if it comes.
2. Choose curl/Stokes/Ampere question if it comes.
3. Choose modified Ampere/Faraday question if it comes.
4. Choose Poynting/polarization if you can write cleanly.
5. Attempt Biot–Savart numerical only if the geometry is clear.

What Not To Do

Avoid these mistakes:

- Do not write only formulas without explanation.
- Do not skip diagrams for curl, boundary condition, Ampere, Faraday and EM waves.
- Do not mix $\mathbf{B}$ and $\mathbf{H}$.
- Do not write $\nabla \cdot \mathbf{B} \neq 0$; it is always zero in classical electromagnetics.
- Do not forget the displacement current term in time-varying Ampere law.
- Do not forget the negative sign in Faraday's law.

Final Minimum Set to Memorise

These are compulsory:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \cdot \mathbf{J} = 0$$

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

6 Final Phase Completion Checklist

Phase	Must Be Able To Do	Status	Remark
0	Know all basic quantities: E, D, H, B, J	Pending	Absolute beginning.
1	Understand vector field, dot product with $d\mathbf{l}$ and $d\mathbf{S}$	Pending	Needed for all derivations.
2	Derive curl in Cartesian coordinates	Pending	Most important.
3	Derive divergence and explain $\nabla \cdot \mathbf{B} = 0$, $\nabla \cdot \mathbf{J} = 0$	Pending	Common short and long answer.
4	Establish Stokes' theorem and $\nabla \times \mathbf{H} = \mathbf{J}$	Pending	Repeated.
5	Use Biot–Savart law and write magnetic vector potential	Pending	Numerical + theory.
6	Derive magnetic boundary conditions	Pending	Long answer.
7	Establish modified Ampere law and Faraday law	Pending	Very high priority.
8	Derive EM wave equation and plane wave solution	Pending	Very high priority.
9	Write Poynting theorem and polarization	Pending	Very high priority.
10	Write complete PYQ answers in semester format	Pending	Final conversion.

Final Conclusion

For Field Theory Part–II, the complete preparation is finished when you can write these five answer families:

Curl–Divergence–Stokes

Ampere–Biot Savart–Boundary Condition

Displacement Current–Modified Ampere–Faraday

EM Wave Equation–Plane Wave–Velocity

Poynting–Polarization–Short Notes

After these, no extra phase is required for Part–II semester preparation.